

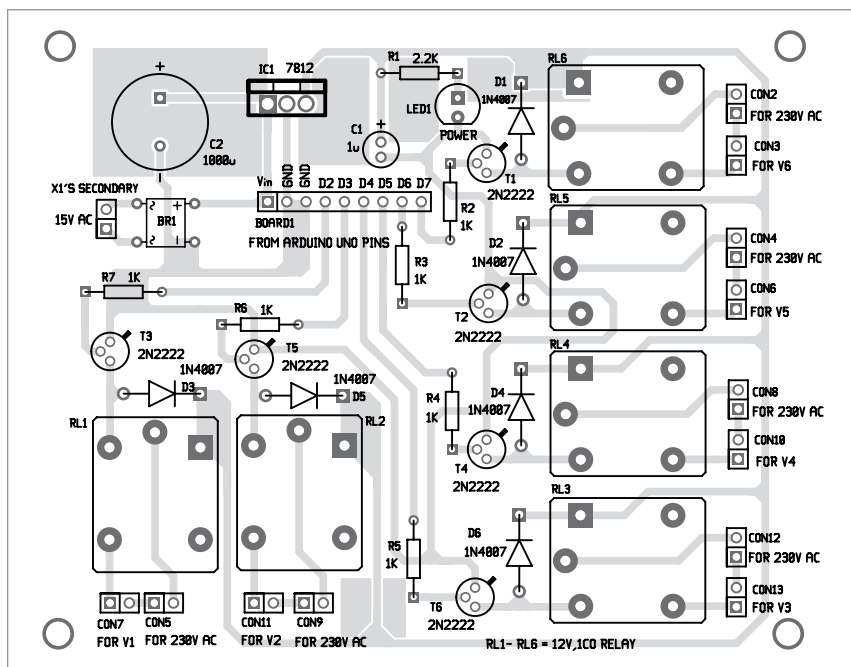


Fig. 6: Component layout of the PCB

PARTS LIST

Semiconductors:

- Board1 - Arduino Uno board
- IC1 - 7812, 12V voltage regulator
- BR1 - 1A bridge rectifier
- D1-D6 - 1N4007 rectifier diode
- T1-T6 - 2N2222 NPN transistor
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

- R1 - 2.2-kilo-ohm
- R2-R7 - 1-kilo-ohm

Capacitors:

- C1 - 1 μ F, 25V electrolytic
- C2 - 1000 μ F, 25V electrolytic

Miscellaneous:

- CON1 - 2-pin connector
- CON2-CON13 - 2-pin terminal connector
- RL1- R6 - 12V, single-changeover relay
- V1-V6 - Pneumatic solenoid valve (2/2 NC)
- X1 - 230V AC primary to 15V, 2A secondary transformer
- DC jack for Arduino Uno
- Heat-sink for IC1



Fig. 7: Author's prototype

volume of the output of that canister is used to flush out nitrogen in the other canister while the remaining part is consumed as oxygen. After the set delay, the relays switch the role of the canisters. This is how pure oxygen is separated from the compressed air.

For better understanding valves' (V1 through V6) and relays' (RL1 through RL6) functioning associated with the project, please refer Table 1. Here, the two canisters are CAN1 and CAN2. The functional block diagram of CAN1 and CAN2 are shown in Fig. 3. The pneumatic arrangement is shown in Fig. 4.

Working of the circuit is simple. First, you need to upload the source code to Arduino and connect the power supply. Relays RL1 through RL6 energise and de-energise according to the software. In the prototype, relays RL1, RL4, and RL5

energised for 40 seconds and relays RL2, RL3, and RL6 de-energised for 40 seconds to open and close the solenoid valves, respectively. Thereafter, the cycle repeats. You can see their status on the serial monitor of Arduino IDE as below:

- Set 2 Pressurised
- Set 1 Exhausting
- Set 1 Pressurised
- Set 2 Exhausting

Construction and testing

An actual-size PCB layout for the oxygen concentrator is shown in Fig. 5 and its component layout in Fig. 6. After assembling the circuit on PCB, connect the 15V secondary across transformer X1. Connect the primary of X1 to 230V, 50Hz AC mains at CON1 as shown in Fig. 1 (not shown in Fig. 6).

Piping is done as shown in Fig. 4. As you can see in Fig. 3 and Fig. 4, there are six solenoid valves (V1 to V6), which are 2/2 normally-closed pneumatic solenoid valves. Inlet for all the solenoid valves is to be referred as port A and outlet as port B. When coil of the solenoid valve is energised, it allows air to flow from port A to port B. Else it will shut the air flow.

An oil-free air compressor should be used to compress the atmospheric air. The pressurised air is supplied to line filter to remove any contaminant